

Cristhian Eduardo Fernández Álvarez

Artificial Intelligence Specialist | Data Scientist | Computer Vision | AWS Cloud

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PROFESSIONAL PROFILE

Electrical engineer with a master's degree in automation, with 5 years of experience in data science and machine learning, and 5 years as a full stack developer. I combine technical skills, analytical vision, and practical experience to design, implement, and optimize artificial intelligence solutions.

In data science and machine learning, my career spans from exploratory analysis and data preparation to the development, validation, and deployment of machine learning and deep learning models in prediction, classification, natural language processing, and computer vision projects. As a full stack developer, I have designed and implemented web applications and comprehensive systems, from architecture to testing and production, seamlessly integrating AI models into production environments, ensuring scalability, performance, and ease of use.

I am passionate about research and innovation, and I keep up to date with the latest advances in AI and automation. I am self-taught, proactive, and results-oriented, with a knack for teamwork and interdisciplinary collaboration to achieve high-impact goals.

TECHNICAL SKILLS

- **Cloud & DevOps:** Amazon Web Services (AWS), infrastructure as code (IaC), deployment and orchestration of production environments.
- **Programming Languages:** Python, R, JavaScript (Node.js, React), PHP.
- **Databases: SQL** (PostgreSQL, MySQL, Oracle, PL/SQL) and NoSQL (MongoDB, DynamoDB).
- **Data Analysis and Science:** Statistical analysis, feature engineering, advanced machine learning techniques for classification, regression, and prediction.
- **Computer vision:** OpenCV, image detection and recognition models.
- **NLP and agents:** LangChain/LangGraph, N8N for applications with LLMs.

EDUCATION

Universidad del Valle

Bogotá, Colombia (Remote)

- Master's degree in engineering with an emphasis on automation.
- Cumulative GPA: 4.6/5
- Research project: "Forecasting the Standardized Precipitation Index (SPI) in the Nariño region using artificial intelligence techniques".
- Electives: optimization of dynamic systems, optimization of linear systems, intelligent control, artificial vision, modern techniques for pattern recognition, and techniques for time series forecasting.

Universidad del Valle

Cali, Colombia

- Electronic Engineer
- Cumulative GPA: 4.19/5
- Project: "Development and Implementation of a Tool for Physical Therapy in Patients with Restricted Mobility in Upper Limbs."
- Electives: Mobile Robotics, Systems Integration for Automation, Programmable Logic Controllers (PLC), User Interfaces and Pattern Recognition.

Colegio INEM Jorge Isaacs

Cali, Colombia

Industrial high school graduate, specializing in Electricity and Electronics

WORK EXPERIENCES

January 2022 – Present

Cali, Colombia

Company: Subocol S.A.

Position: Artificial Intelligence Specialist.

Responsibilities:

- Development and application of AI models for the auto insurance sector.
- Improvement and maintenance of existing AI services and models.
- Research and development of AI models.
- Statistical data analysis.
- Implementation of web services.
- Management of SQL and NoSQL databases.
- Development of ETLs.
- Data visualization.
- Management of services on AWS and platforms for MLOps.

May 2016 - February 2022

Cali, Colombia

Company: Ipinnovatech LTDA.

Position: Senior Development Engineer.

Responsibilities:

- Design and implementation of responsive web applications for managing and querying different SQL/NoSQL databases and third-party services.
- Data visualization.
- Development of RETAIL solutions.
- Implementation of web services.
- Server management.
- Research in artificial intelligence using convolutional neural networks (CNNs) for object detection in images.
- Implementation of computer vision algorithms.
- Management of AWS services.

February 2016 - March 2017

Cali, Colombia

Company: Universidad del Valle.

Position: Research assistant monitor.

Responsibilities:

- Implementation of computer vision algorithms.
- Development of routines for the movements of the NAO humanoid mobile robot using Python. The following links show two test demonstrations of the NAO robot's behavior when imitating human movements through the use of a stereo camera.
<https://www.youtube.com/watch?v=Vgwi0LyQMCg>
<https://www.youtube.com/watch?v=NT2LTXGOLJE>